

## RND-6: SAFE IN-TIME COVARIATES ??? CUPAC-STYLE ????????

> ??????: ? ??????. ??????????? ?????????????? ????????? ? ??????? ???????  
> A/B ? CUPAC A ? A+B ? A+B+C; ???????? ????????? open-source ?????????? ? ?????????? ???.  
> ????????? ??????? ?? ??????? ? notebook.ipynb (??????????, seed=11).

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?????????, ??? ? ? ?????????????????? ?????????? ?????????? in-time ?????????? ??????  
scikit-CUPAC ?????????????????? ?????????? ?????????? (? ?????????????? ?????????? ?????????) ???  
????????????? ?????????????????????? ?????????????????? ??????? ATE.

??????? ?????????? (SAFETY LAYER)

? **A** pre-treatment ? ??? ? CUPAC;  
? **B** expert-safe in-time ? ????????? second-stage ??????????  
? **C** balance-gated in-time ? ?????????? ?????????? ????? balance/missingness gate;  
? **D** dag-required ? ??? estimator (????? causal-?????????);  
? **E** mediator-risk, **F** leakage ? ?????????? ??? ?????????? ?????? ATE;  
? **unsafe\_demo** ? ?????? ?????????????? ??????, ?? ??????????

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ab\_hypex (A/B baseline) • hypex\_cupac (reference/parity, ??? ???????) •  
sklearn\_cupac\_A ? sklearn\_cupac\_A\_plus\_B\_linear ? sklearn\_cupac\_A\_plus\_B\_plus\_C\_linear  
(????????? ?????????? ??????????) • unsafe\_demo\_optional (??????????????).

????????????? ?? ?????????? (N=8000, SEED=11, ?????????? ATE = 2.8)

| method                               | ate   | p_value | variance_reduction_vs_ab_% | incremental_vs_predecessor_% | safety_status  |
|--------------------------------------|-------|---------|----------------------------|------------------------------|----------------|
| ---                                  | ---   | ---     | ---                        | ---                          |                |
| ab_hypex                             | 2.758 | 0.0     | 0.0                        | ?                            | ok             |
| hypex_cupac                          | 2.785 | 0.0     | 13.56                      | ?                            | reference_only |
| sklearn_cupac_A                      | 2.801 | 0.0     | 25.63                      | 25.63                        | ok             |
| sklearn_cupac_A_plus_B_linear        | 2.775 | 0.0     | 42.38                      | 22.52                        | ok             |
| sklearn_cupac_A_plus_B_plus_C_linear | 2.766 | 0.0     | 52.80                      | 18.07                        | ok             |
| unsafe_demo_optional                 | 1.000 | 0.0     | 83.17                      | ?                            | unsafe_demo    |

(????????? ?????????? ????????? ? ??????? ?????????????? ?????????? ?? ?????????? ??????????)

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1. ?????? ?????????????? ??????? ??? ? ?????????????????? ?????????: A (~26%) ? +B (~42%) ? +B+C (~53%)  
????????? ?????????? ?????????????? A/B, ??? ??? ATE ?????????? ?????????????? (? 2.8).
2. CUPAC parity: hypex\_cupac (~14%) ?????????? ?? ?????? ? sklearn\_cupac\_A;  
????????????? ?????????????? ??????? ??? R&D ? ?????????????????? ??? ??????????
3. Balance/missingness gate ?????????????? A/B/C (|SMD|?0) ? ?????????? D/E/F ? unsafe\_demo  
(|SMD| ? 0.45?1.14, p?0) ? ?????????, ?????? ?? ?????? ?????? ? estimator.
4. unsafe\_demo ??? ? ?????????????????? «????????? ??????????» (~83%), ?? ?????????? ??????? ATE

?????? ?? ????????? 2.8 ? ????????? ? 1.0 (??, ??????) ? ????????????? ??????  
outcome-derived ?????????; ? ????????????? ?????? ?? ??????.

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? balance-test ? ????????????? gate (????? ????????????? ?????????? E/F), ? \*\*??\*\* ?????????????????  
???????????? ?????????????; ??? D-???????????? ?????? ????????? DAG/causal-?????  
? E/F ?????????; safe-intime ????????????? \*\*??????\*\* ? ????????? second-stage, ?? ?  
???????????????? ML-??????.

?????? ?????????? ?? ?????????? ??????

? \*\*????????\*\* ????????????? ?????? ??????? \*\*A+B+C\*\* (???????? ????): ?????? ??? ??????  
A/B/C/D/E/F ????????? ?? ?????????????.  
? \*\*????????? ?????????? ?????????? ?????? ?????????????? ? ????????? A-only\*\* CUPAC-?????????  
????? ????? (download + schema\_summary) ?????? Orange Belgium, Lenta, Criteo, X5  
RetailHero,  
MegaFon, Lazada/DESCN, CUVET-policy, MT-LIFT, Dunnhumby ? semi-synthetic causal-?????????  
? \*\*Hillstrom\*\* ?????????? pipeline/A-only, \*\*??\*\* B/C.  
? ????????? safe in-time \*\*B/C\*\* ??????????: ????????? (‘f’, ‘PC’, ‘X\_’),  
?????  
??? ???-????????? ????????? (A), ??? ???-treatment ????????? (E/F). ????????? ? ??????????  
????????????????? ??? ?????????? ??? ????????? (X5), ??? ?? ????????????????? (Dunnhumby).  
? \*\*??????? A+B+C-????????? ?? ?????????? ?????????? ?????????? ?? ?????????????\*\* ? ?????????  
????????????????, ??? ?????? ?????? ?? ?????? ?????????? B/C-????????? ? timing-evidence.  
? ?????? ? ?????????????? ?????????????? ??????????: \*\*docs/06\_real\_dataset\_raw\_audit.md\*\*.

???????????????? ? ?????????? ? HYPEX

?????? ????????????? ????????????? ?????????? ?? ?????????? ? ????????????? ? HypEx CUPAC. ?????  
?????????  
? HypEx ??????????: (1) ????????????? ?????? B/C ?? ????????????? ?????????? open-source ??????????  
(2) ????????????? balance/missingness gate ??? ?????????????????? ??????????; (3) ?????????????  
?????? E/F ? ?????????? ?????? unsafe\_demo. ????????????? ?????????????????? ????????????? CUPAC ??  
????????? ?????? ? VarWar (????????? ?????????? ~34% ? ~33%; ?????? ?????????, ?? ??????????).

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bash  
pip install -e .[ml]  
??. NOTEBOOK.IPYNB; ?????? ATE-CI ? ?????? ?????????????? ?? RND\_REPORTS.BENCHMARK  
  
python tools/generate\_pdf.py # ?????????????? report.pdf